

KARAN KAPUR

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Education

University of Wisconsin - Madison

Sep. 2022 – May 2026

Bachelor of Science in Computer Science, Data Science, Minor in Entrepreneurship

Madison, WI

Technical Skills

Languages: Python, Java, C, C#, C++, SQL, TypeScript, JavaScript, HTML/CSS, R

Developer Tools: Git, Docker, UNIX/Linux, AWS (Bedrock/S3), VS Code, Google Cloud, Android Studio, RStudio, Unity, Figma, Tableau, Redis

Technologies/Frameworks: PyTorch, PyRadiomics, TotalSegmentator, Node.js, React.js, Google Firebase, OAuth 2.0, Flask, Selenium, Pandas, NumPy, ggplot2, Scikit-Learn, Matplotlib, LangChain, Pinecone, Streamlit, Apache Spark, Apache Kafka, Apache Cassandra, HDFS, BigQuery, CUDA/PyO3 FFI, Hugging Face, gRPC, Raft, LSM-trees

Experience

Thaw (Open Source Library)

April 2026 – Present

Co-Founder & Cloud Storage Engineer

Madison, WI

- Engineered thaw, a Rust + CUDA inference system that reduced vLLM session branching from $\sim 340s$ cold boot to 0.88s median fork time on H100s by snapshotting & restoring live LLM state.
- Owned the cloud storage layer, enabling durable storage & retrieval of large inference snapshots by designing persistence workflows for weights, KV cache, prefix-cache metadata, & scheduler state.
- Improved fork reliability & restore readiness by integrating cloud-backed snapshot storage with thaw's restore pipeline, allowing child sessions to resume from saved parent state instead of repeating prefill.
- Supported high-throughput model restoration by connecting storage workflows to thaw's double-buffered restore path, contributing to 14.3 GB/s weight restore throughput & $3.4\times$ faster 70B model loading. (Co-Founded with N. Matteson & M. Yu)

Department of Electrical & Computer Engineering, UW - Madison

January 2025 – May 2026

Computer Support Assistant

Madison, WI

- Provided front-line technical support across **1,500+** university-managed Windows, macOS, & Linux machines, troubleshooting hardware, software, account, network, & system issues for faculty, staff, researchers, & students.
- Reduced device downtime by **40%** by developing automated monitoring & log-analysis scripts to identify recurring system failures, performance issues, & support bottlenecks.
- Analyzed & resolved **1,000+** IT tickets within the department's ITSM platform, improving triage workflows, prioritization, & response consistency while helping raise end-user satisfaction to **98%**.

Eurofins Scientific

Jan 2026 – May 2026

Data Engineer (UW-Madison Senior Design Capstone Project)

Madison, WI

- Architected end-to-end predictive analytics system using FastAPI, React, & TypeScript to transform **10K+** historical analytical records into pre-test analyte forecasts, streamlining technician workflows & reducing rerun-related operational delays.
- Engineered scalable data ingestion & preprocessing workflows across laboratory datasets for **4+** business unit, standardizing noisy chemistry data into prediction-ready inputs for improving reliability of downstream forecasting.
- Utilized historical sample similarity and statistical modeling to generate analyte estimates, confidence scores, & comparable sample recommendations, enabling faster & more informed lab decision-making.

Wisconsin Summer AI Laboratory - OpenAI x UW-Madison

June 2025 – August 2025

AI Fellow

Madison, WI

- Built a retrieval-augmented platform serving **10+** concurrent users with LangChain, Pinecone vector search, & both OpenAI & Hugging Face model APIs for domain-specific document retrieval at scale, reducing manual document lookup time by **40%**.
- Deployed a Streamlit web app enabling real-time semantic search & context-aware inference for non-technical users.
- Designed & implemented multi-domain scaling strategies enabling handling of **50+** concurrent queries with **sub-2s** latency, reducing infrastructure costs by **25%**.

HCLSoftware

July 2024 – September 2024

Automation & Compliance Engineering Intern

Noida, India

- Cut regression testing cycles by **60%** by authoring **50+** automated test scripts for a large public-sector SaaS platform.
- Authored compliance validation reports aligning the platform with government IT standards & audit requirements.
- Improved onboarding conversion by **30%** via UX audits & analytics-driven workflow redesign with cross-functional teams.

Personal Projects

NFL Quarterback Decision Metric | *Python, R, Spatial Modeling*

December 2025

- Designed outcome-independent QB decision quality metric using receiver-defender spatial separation; validated on **5** seasons of game film with **85%** correlation to professional QB performance outcomes
- Modeled receiver-defender separation using player-tracking kinematics at ball arrival and delivered insights applicable to scouting, offensive scheme design, and player valuation.

WSH - Custom UNIX Shell | *C, POSIX, Linux*

October 2025

- Implemented multi-stage pipeline execution with fork, execv, dup2, and UNIX pipes, enabling concurrent inter-process communication between **120+** chained commands and proper process control.
- Developed built-in commands, PATH resolution, and dynamic data structures (hash maps for aliases, dynamic arrays for history) to support flexible command execution and persistent shell state.

MLB Pitching Release Point Analysis | *Python, R, MLBStatcast*

September 2025

- Analyzed release-point data for 700+ MLB pitchers to identify mechanics linked to run prevention by combining Statcast & Baseballr data with pitcher-level performance metrics.
- Identified a ~ 0.45 ERA advantage for specific release profiles by building regression & clustering models that linked horizontal release point, vertical release height, & release depth to pitching outcomes.
- Surfaced performance patterns beyond traditional handedness splits by segmenting pitchers into release-mechanics clusters based on arm slot, release height, & extension.
- Translated biomechanical trends into scouting & player-development insights by recommending higher & farther release profiles as potential indicators for pitcher evaluation.
- Designed the analysis for future model expansion by structuring the workflow to incorporate spin rate, pitch movement, pitch type, & velocity features for deeper pitcher performance evaluation.

Formula 1 Race Strategy Prediction Pipeline | *Python, FastF1, Visual Crossing API, Random Forest*

August 2025

- Built a real-time race-outcome prediction pipeline that fused live F1 telemetry, pit-stop strategy, & weather data into a single unified model; achieving **<5%** Mean Absolute Error & **sub-100ms** inference latency on live feeds.
- Delivered live pit-stop timing recommendations with 92% confidence.

AI-Powered Intrusion Detection System | *Python, Scikit-learn, CICIDS 2017*

June 2025

- Designed a real-time packet inspection pipeline aligned with enterprise cybersecurity workflows.
- Trained ML Models on **1M+** labeled CICIDS 2017 records across **80+** network features, achieving **90%** F1-score and 94% precision for threat detection; deployed in production intrusion detection pipeline protecting **500+** enterprise endpoints.

Leadership / Extracurricular

Kappa Eta Kappa

May 2024 – December 2024

President

University of Wisconsin - Madison

- Managed executive board of **8** members and ran weekly meetings to oversee progress in essential parts of the chapter.
- Led chapter of **50+** members to work towards goals that improve and promote community service, academics, and unity.
- Enhanced chapter reputation & networking reach by representing the organization at national councils & alumni boards.